

Topic: Microinstruction Sequencing and Execution

Q. Explain the concepts of Microinstruction Sequencing and Execution in a Microprogrammed Control Unit. What is Control Memory

[DEF] Definition to Remember

In a Microprogrammed Control Unit, a single machine instruction is executed by running a series of smaller steps called Microinstructions (collectively forming a Microprogram). These microprograms are stored in a dedicated ROM called Control Memory.

Microinstruction Sequencing & Execution

* It is a fast, Read-Only Memory (ROM) located inside the CPU, completely separate from main system memory.

2. Microinstruction Execution

A microinstruction is basically a control word. The bits in this word directly translate to physical control lines in the CPU.

3. Microinstruction Sequencing

Sequencing is the process of determining the address of the *next* microinstruction to fetch from Control Memory. The hardware unit responsible is the Sequencer.

4. Workflow Loop

1. Map Machine Instruction Opcode -> Control Memory starting address.

2. Se

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Microprogram: A sequence of microinstructions that execute one machine instruction.
2. Control Memory: A dedicated, fast ROM inside the CPU used solely to store microprograms.
3. Execution: A microinstruction acts as a control word; its bits directly activate CPU control lines (gates/ALU).
4. Sequencing: The process of finding the address of the next microinstruction.
5. The Sequencer determines the next address.
6. Sequencing methods: Next Address Field, Conditional Branching (based on flags), Opcode Mapping, and Subroutines.
7. The execution is a loop of fetching, signaling hardware, and determining the next micro-address until finished.

[RECALL] Quick Recall

Microprogram (steps for 1 instruction) -> Control Memory (ROM storing steps) -> Execution (bits turn into hardware signals) -> Sequencing (finding next step via Opcode/Next Field/Branching)